2. Requirements Gathering

Project: NutriVision — AI-Driven Dietary Health Assistant
Document status: Initial draft for validation

2.1 Stakeholder Analysis

Stakeholder	Interest	Needs	Influence	Engagement
End users	High	Simple, useful, understandable dietary guidance	High	Usability feedback and scenario testing
Project team	High	Clear scope, manageable tasks, reliable tools	High	Weekly planning and pull-request review
Team leader	High	Delivery visibility and consistent integration	High	Track milestones, risks, and decisions
Instructor	High	Evidence of planning, implementation, testing, and learning	High	Milestone reviews and feedback
API/data providers	Medium	Valid requests and compliant usage	Medium	Integration documentation and error monitoring
Firebase/hosting services	Medium	Correct configuration and secure access	Medium	Security rules and service monitoring
Academic evaluators	High	Clear documentation and demonstrable outcomes	High	Final report, demo, and presentation

2.2 User Personas

Persona A — Everyday meal planner

An adult or student who wants to understand a meal quickly and find an easier healthier alternative without needing detailed nutrition knowledge.

Goals: quick insight, practical suggestions, simple history.
Pain points: confusing nutrition information, limited time, inconsistent tracking.

Persona B — User with dietary restrictions

A user who wants recipe suggestions filtered by preferences or restrictions such as vegetarian, low-sugar, or allergy-related exclusions.

Goals: safer suggestions, transparent ingredients, personalized recipes.
Pain points: unsuitable recommendations and difficulty checking ingredients.

Persona C — Project evaluator

An instructor or evaluator who needs to see a clear user journey, sound requirements, stable implementation, testing evidence, and responsible handling of health-related information.

Goals: assess quality, relevance, usability, and technical decisions.
Pain points: unclear scope, unsupported claims, missing test evidence.

2.3 User Stories

ID	User story	Acceptance criteria
US-01	As a user, I want to create or configure my profile so that suggestions reflect my preferences.	Preferences can be saved, viewed, and updated.
US-02	As a user, I want to record a meal description so that I can analyze it.	A valid meal entry can be submitted; empty input is rejected with guidance.
US-03	As a user, I want to see nutritional insights so that I can understand my meal better.	The result screen shows available nutrition data, source/status, loading, and error states.
US-04	As a user, I want to receive a healthier alternative so that I can make a practical change.	At least one alternative is shown when the service returns one; unavailable results are handled clearly.
US-05	As a user, I want recipe suggestions that match my restrictions so that I can prepare suitable meals.	Suggestions respect the saved filters and show ingredients/instructions returned by the service.
US-06	As a user, I want to save my meal so that I can review it later.	A successful save appears in meal history and remains available after reopening the app.
US-07	As a user, I want to view my meal history and basic charts so that I can notice patterns.	History supports loading, empty, populated, and failure states.
US-08	As a user, I want my meal logs synchronized across devices so that my information stays consistent.	Authorized changes are reflected through Firestore synchronization.
US-09	As a user, I want clear explanations and a health disclaimer so that I understand the limits of the advice.	Disclaimer is visible at an appropriate point in the experience.
US-10	As a team member, I want testable services and widgets so that changes can be reviewed safely.	Critical logic and key UI states have automated tests.

2.4 Main Use Cases

Use case	Actor	Main flow	Alternative/error flows
UC-01 Configure profile	User	Open profile → choose preferences/restrictions → save	Invalid or incomplete data is explained
UC-02 Analyze a meal	User	Enter meal → submit → view nutrition result	Empty input, timeout, invalid response, or unavailable service
UC-03 Find an alternative	User	Open result → request alternative → review suggestion	No suitable alternative is returned
UC-04 Generate a recipe	User	Select preference → request recipe → view ingredients and steps	Unsupported restriction or service failure
UC-05 Save meal log	User	Confirm meal → save → receive confirmation	Authentication or Firestore write failure
UC-06 Review history	User	Open history → browse entries → view details/chart	Empty history or read failure
UC-07 Synchronize data	Authorized user	Update meal log → Firestore listener receives change	Offline state, conflict, or permission denial

2.5 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow a user to create or update dietary preferences.	Must
FR-02	The system shall validate required meal input before sending a request.	Must
FR-03	The system shall send meal-analysis requests through a dedicated networking layer.	Must
FR-04	The system shall display loading, success, empty, and failure states for remote requests.	Must
FR-05	The system shall display the available nutritional results in a readable format.	Must
FR-06	The system shall request and display healthier alternatives when available.	Should
FR-07	The system shall generate or retrieve recipe suggestions using the selected restrictions/preferences.	Must
FR-08	The system shall allow an authorized user to save a meal record.	Must
FR-09	The system shall display saved meals as a history list.	Must
FR-10	The system shall display basic nutritional summaries or charts based on available records.	Should
FR-11	The system shall synchronize authorized meal records through Firestore.	Must
FR-12	The system shall prevent unauthorized users from reading or modifying another user's records.	Must
FR-13	The system shall show a clear disclaimer that the output is informational and not medical diagnosis.	Must
FR-14	The system shall handle network, parsing, authentication, and database errors without crashing.	Must

2.6 Non-functional Requirements

Category	Requirement	Verification
Usability	Core actions should be understandable without technical knowledge.	Usability review and test scenarios
Responsiveness	Layouts shall adapt to supported phone screen sizes and orientations as agreed by the team.	Widget tests and manual device tests
Performance	The app shall show progress for remote work and avoid blocking the UI thread.	Manual profiling and request tests
Reliability	Expected service failures shall produce recoverable UI states.	Negative test cases
Security	Secrets shall remain outside Git; data access shall follow least privilege.	Repository audit and Firebase rules tests
Privacy	The app shall collect only data required for the agreed features and explain its use.	Privacy checklist and review
Maintainability	Networking, data, business logic, and presentation concerns shall be separated where practical.	Code review
Testability	Critical calculations, parsing, and UI states shall be independently testable.	Unit/widget test report
Compatibility	The MVP shall run on the agreed Flutter-supported Android target; other platforms are documented separately.	Build and device test
Accessibility	Text, controls, contrast, and error messages should be readable and understandable.	Accessibility checklist

2.7 Data and Privacy Requirements

- Meal records and preferences must be associated with the correct authorized user.
- Firestore Security Rules must reject unauthorized reads and writes.
- API responses must be validated before being shown to the user.
- Logs must not expose tokens, passwords, private meal data, or personal identifiers.
- The team must agree on retention and deletion behavior before final implementation.
- The application must not present AI output as a medical diagnosis or guaranteed treatment.

2.8 Assumptions and Open Questions

- The final AI provider and nutrition-data source are not confirmed yet.
- Authentication requirements need confirmation before the Firestore schema is finalized.
- The exact target Android versions and supported devices are TBD.
- The final chart metrics depend on the nutrition data returned by the chosen source.
- Dietary restrictions and allergy handling need a final approved list.
- The team must decide whether image-based meal recognition is part of the MVP or a future enhancement.